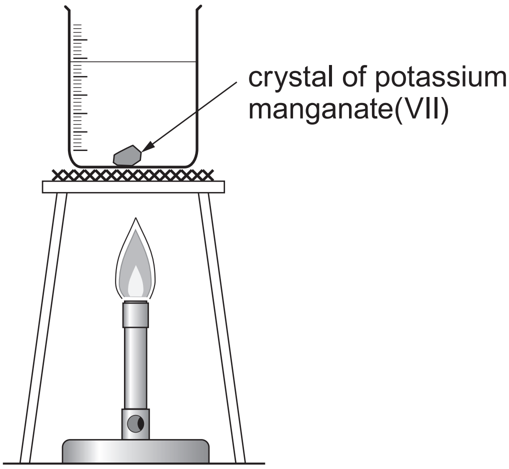
Examiner only

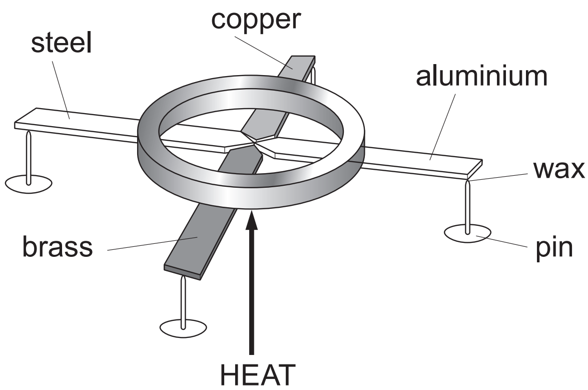
2. Heat can be transferred through materials by conduction, convection and radiation. Three groups of students carried out two experiments to show different methods of heat transfer.

They used the following equipment to perform the two experiments.

<table>
<tr><th>Experiment 1</th><th>Experiment 2</th></tr>
<tr><td>Experiment to show heat transfer through the liquid by ....



</td><td>Experiment to show heat transfer through metals by ....



</td></tr>
</table>

(a) **Complete the diagrams** above by identifying the type of heat transfer shown by each experiment. [2]

(b) The results of **Experiment 2** are shown in the table below.

Metal	Time taken for the pin to drop off (s)			
	Group 1	Group 2	Group 3	Mean
Steel	73	78	74	....
Copper	14	15	13	14
Aluminium	20	19	18	19
Brass	35	37	46	36

(i) **Complete** the table. [1]

(ii) **Circle** the anomalous result. [1]

Examiner only

(iii) Use the results to arrange the metals in order starting with the one that transfers heat the best. [1]

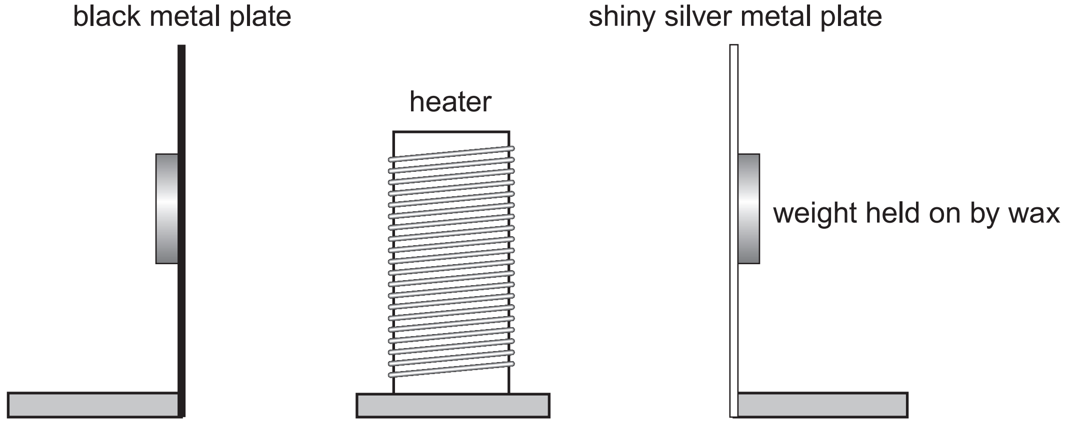
Best metal for transferring heat ....

....

brass

Worst metal for transferring heat ....

(c) The students carried out another experiment as shown in the diagram below.



Explain why this experiment shows that the plates are heated by radiation and not by convection or conduction. [3]

....

....

....

3445U101 07

8